

Strategic Patrol Optimization Model

1. Sets and Indices

- P : Set of all precincts in Brooklyn.
- S : Set of 4-hour time-indexed shift windows.
- $p \in P$: Index for a specific precinct.
- $s \in S$: Index for a specific shift window.

2. Parameters

- $\lambda_{p,s}$: Predicted incident volume for precinct p during shift s .
- $w_{p,s}$: Priority penalty weight (2.0 for high-risk windows, 1.0 otherwise).
- B : Total daily officer-shift budget.
- ϵ : Officer coverage efficiency constant (0.05).
- L : Safety floor / Minimum units per precinct ($0 \leq L \leq 2$).
- U : Saturation cap / Maximum units per precinct.

3. Decision Variables

- $x_{p,s} \in \mathbb{Z}^+$: Number of officers assigned to precinct p during shift s .
- $r_{p,s}$: Residual (uncovered) crime risk in precinct p during shift s .

4. Objective Function

Minimize the total weighted residual risk across all precincts and shifts:

$$\text{Minimize } Z = \sum_{p \in P} \sum_{s \in S} (w_{p,s} \cdot r_{p,s}) \quad (1)$$

5. Constraints

Daily Workforce Capacity

$$\sum_{p \in P} \sum_{s \in S} x_{p,s} \leq B \quad (2)$$

Risk Mitigation Logic

$$r_{p,s} \geq \lambda_{p,s} - (x_{p,s} \cdot \epsilon) \quad \forall p, s \quad (3)$$

Safety Floor and Saturation Cap

$$L \leq x_{p,s} \leq U \quad \forall p, s \quad (4)$$

Non-Negativity Constraints

$$x_{p,s} \geq 0 \quad \text{and} \quad x_{p,s} \in \mathbb{Z} \quad \forall p, s \quad (5)$$

$$r_{p,s} \geq 0 \quad \forall p, s \quad (6)$$